

Numerical PDEs Homework #1

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Problem 1: Solve Stokes

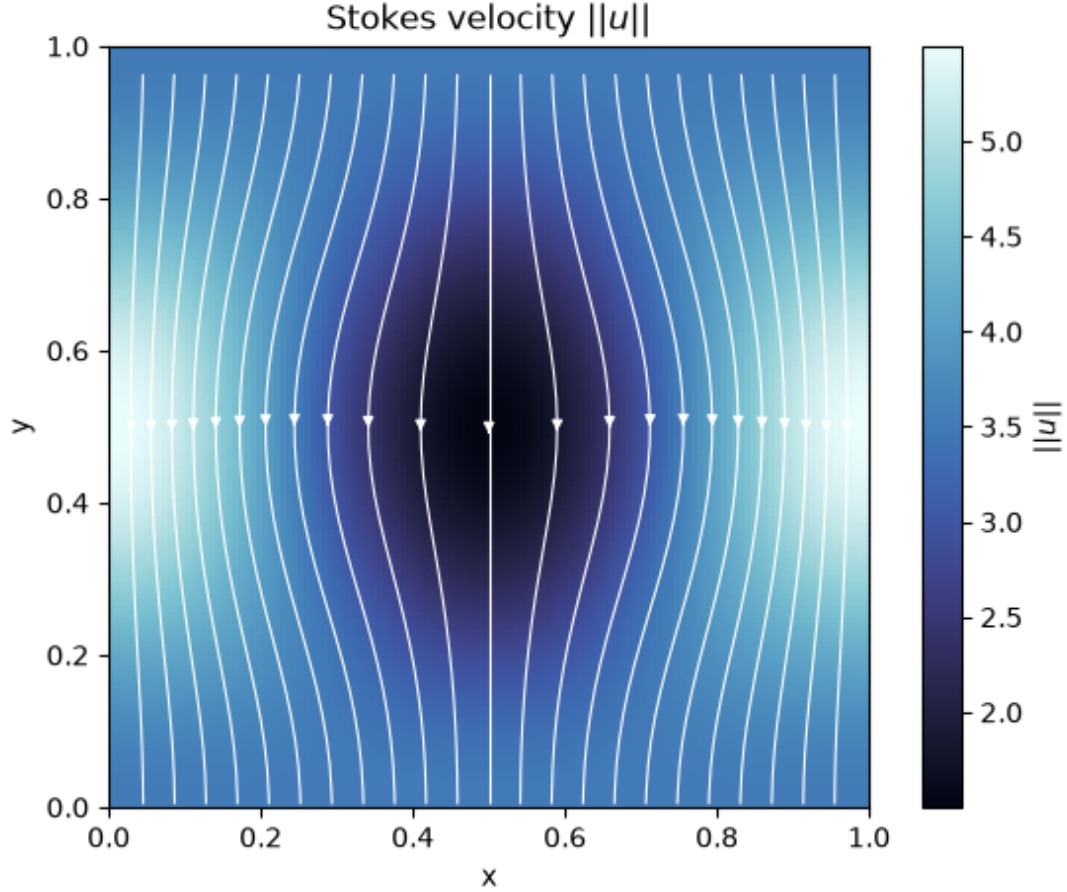


Figure 1: Generated solution plot of the magnitude of $\mathbf{u}$.

I implemented this manufactured solution method as described with a staggered grid and the specified boundary conditions. I used python with numpy and scipy to build the matrices, and used spsolve to solve the resulting matrix equation. As noted in the assignment, I had to pin one row of P to 0 to get a unique solution, and subtracted off the mean for the error computations in the following problem. The generated solution plot in Figure 1 matches the example solution plot included in the assignment.

Problem 2: Error Analysis

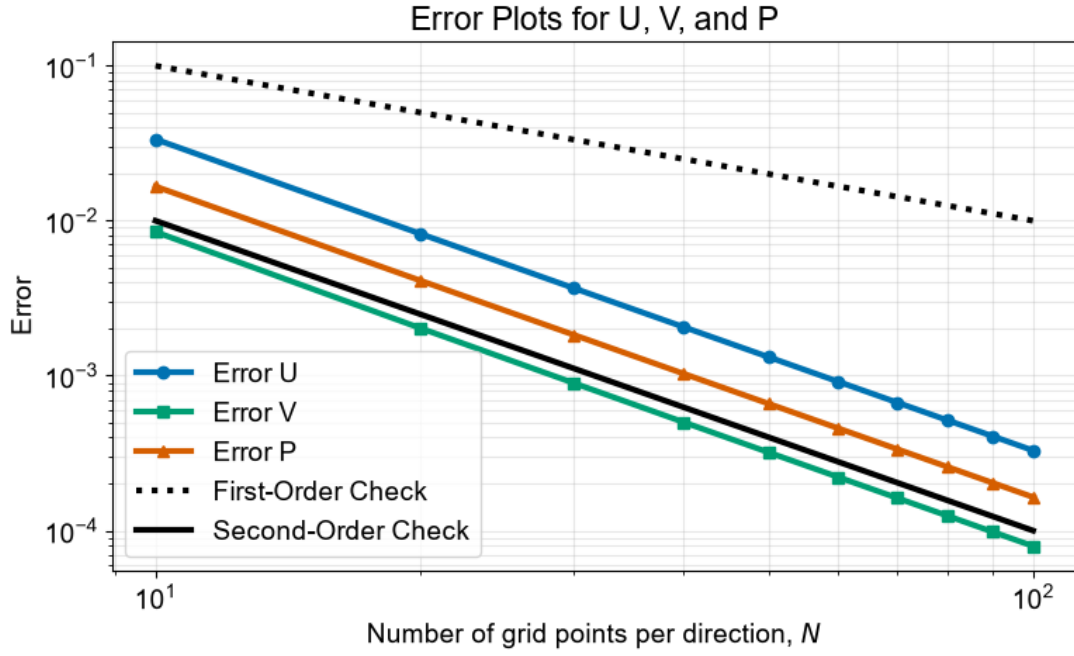


Figure 2: Error plots over increasing N with first and second order convergence plots.

Figure 2 shows that increasing the number of grid points in each directions results in a second order decrease in error, which is expected since centered differences' errors scale as $\mathcal{O}(h^2)$. Again, this plot matches the example plot included in the assignment.

Problem 3: Charlie's Rule

I modified my stokes solver function to take a function for the V boundary condition instead of a constant value. This allows the simulation of varying inlet and outlet velocity across the horizontal boundaries.

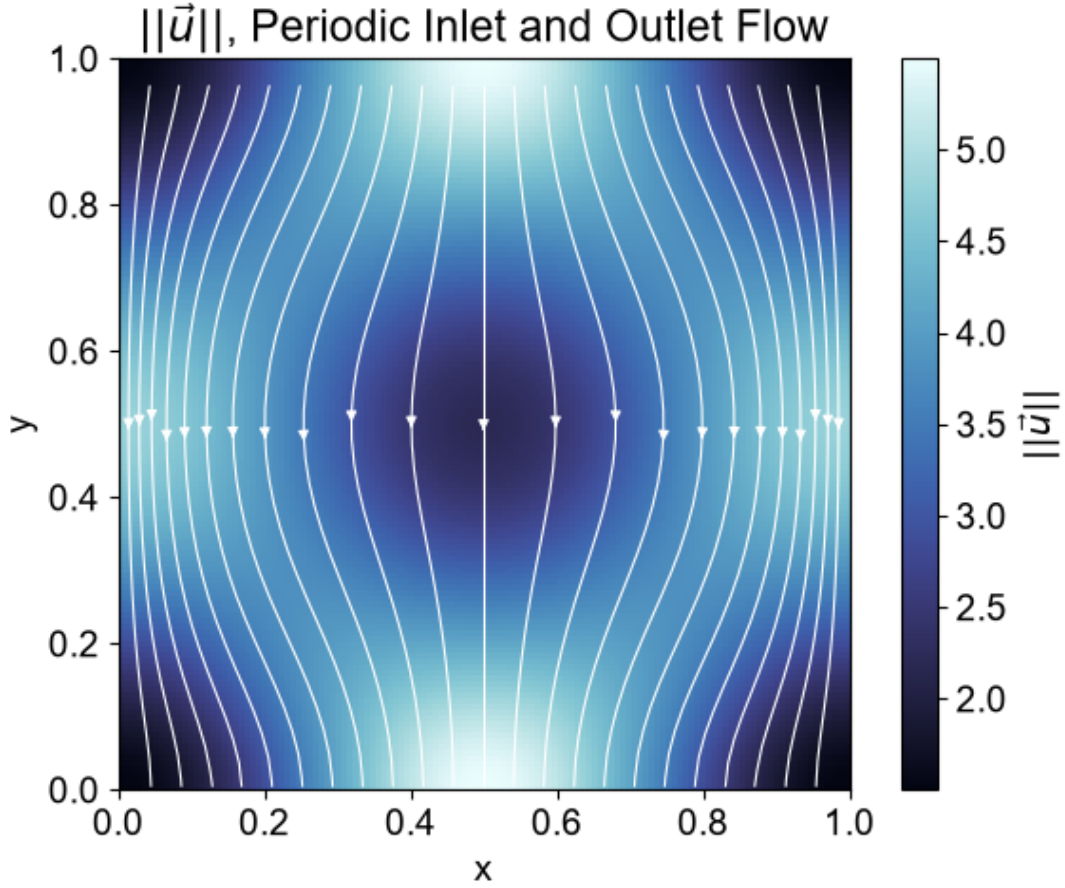


Figure 3: Solution with a varying boundary value, $V_{bndry} = -3.5 + 2 * \cos(2\pi x)$

Figure 3 shows the velocity magnitude plot with a periodic function for the V boundary values on top and bottom. This results in higher magnitude velocity in the centers and lower on the left and right boundaries, but still simulates an incompressible fluid since the inlet and outlet velocities are the same.

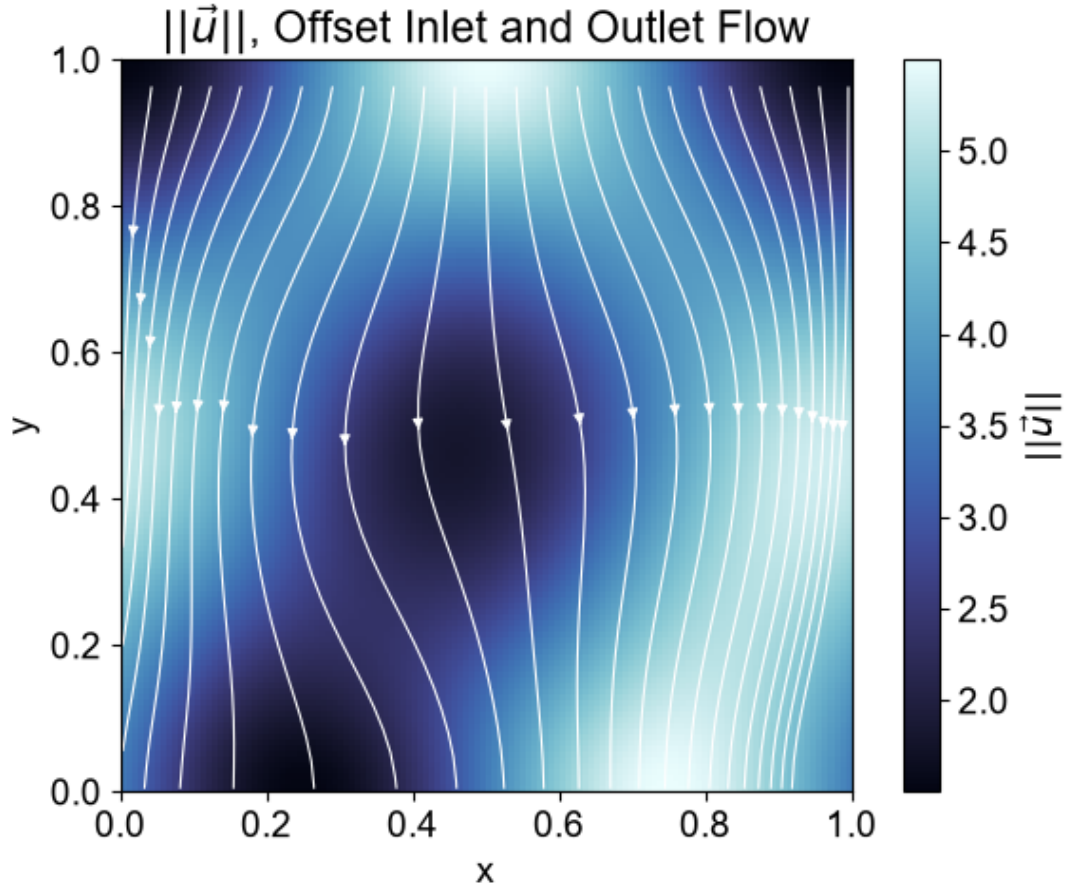


Figure 4: Solution with offset varying boundary values, $V_{top} = -3.5 + 2 * \cos(2\pi x)$ and $V_{bottom} = -3.5 + 2 * \sin(2\pi x)$.

We can also simulate offset varying boundary values, where the total inlet and outlet flows still match but their periods are offset, as seen in Figure 4. This creates a more interesting geometric pattern, but is still valid under the constraints of this simulation.

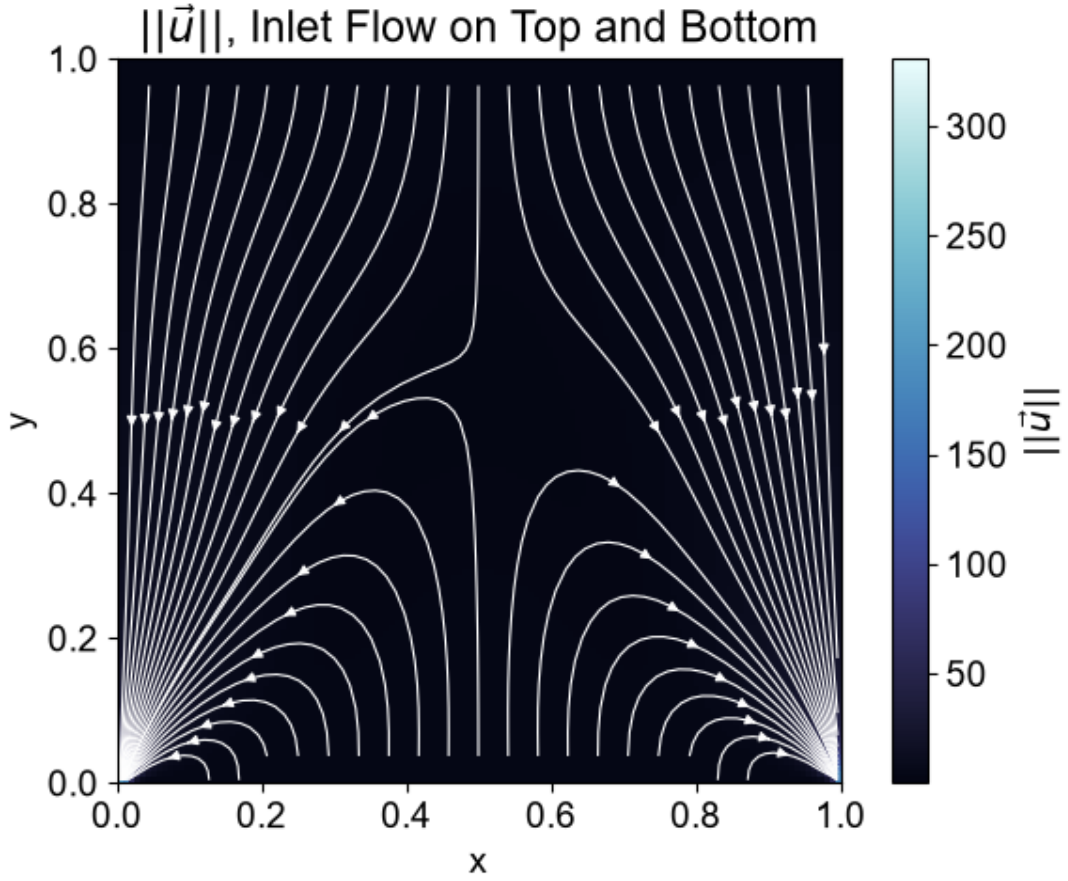


Figure 5: Solution with equal but opposite inlet flow on the top and bottom boundaries.

Figure 5 shows the issue with setting the top and bottom boundaries to opposite values. With the the top set to -3.5 and the bottom set to 3.5 , and periodic boundary conditions enforced on the vertical boundaries, there is nowhere for the inlet flow to escape. The resulting plot shows that the current solver is incompatible with these boundary conditions and would need a horizontal outlet, a sink, or a compressible fluid model.